Log Analysis (Splunk) – Firewall, VPN, IDS
Network Essentials/Investigating the Breach

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Overview
This project focuses on basic log analysis using Splunk
The goal was to identify suspicious activity across multiple log sources and correlate events.

Data Source:
- Firewall Logs (1284) – network connections (allowed/blocked)
- IDS Logs (983) – detected suspicious or malicious activity
- VPN Logs (220) – authentication and assigned internal IP addresses

In total, 2487 events were analyzed covering the period: August 25th, 2025 – September 30th, 2025

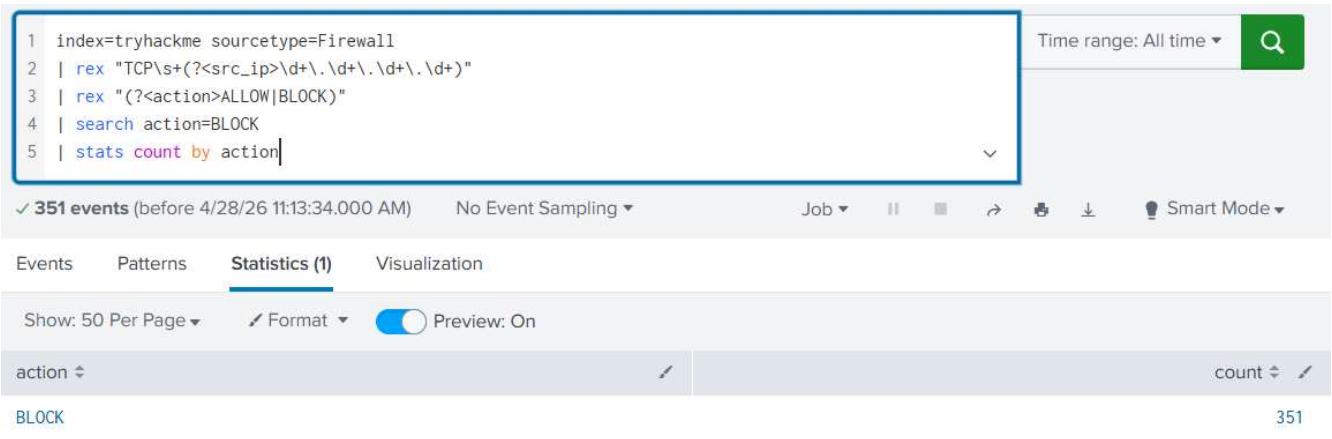
Methodology:
1. identify suspicious IP addresses in firewall logs
2. Correlate them with VPN activity
3. Analyze IDS alerts to understand the type of activity

Due to unparsed logs, simple Splunk queries and field extraction (rex) were used.

Findings:

1. Firewall Analysis

A total of 351 events were blocked by the firewall.



The following IP addresses were identified as suspicious:

- 203.0.113.45
- 203.0.113.10
- 10.8.0.23 (internal/VPN-assigned IP)

```
index=tryhackme sourcetype=Firewall
| rex "TCP\s+(?<src_ip>\d+\.\d+\.\d+\.\d+)"
| rex "(?<action>ALLOW|BLOCK)"
| search action=BLOCK
| stats count by src_ip
```

Time range: All time

✓ **351 events** (before 4/28/26 11:14:48.000 AM) No Event Sampling Job Smart Mode

Events Patterns **Statistics (3)** Visualization

Show: 50 Per Page Format Preview: On

src_ip	count
203.0.113.45	279
203.0.113.10	46
10.8.0.23	26

a) **IP 203.0.113.45** – total events: 297 → 279 Blocked & 18 Allowed

New Search Save As Create Table View Close

```
index=tryhackme sourcetype=Firewall
| rex "TCP\s+(?<src_ip>\d+\.\d+\.\d+\.\d+)"
| rex "(?<action>ALLOW|BLOCK)"
| search src_ip="203.0.113.45"
| stats count by action
```

Time range: All time

✓ **297 events** (before 4/28/26 11:17:28.000 AM) No Event Sampling Job Smart Mode

Events Patterns **Statistics (2)** Visualization

Show: 50 Per Page Format Preview: On

action	count
ALLOW	18
BLOCK	279

Those blocked requests tried to access multiple ports:

- 443 (HTTPS), 4444, 23 (Telnet), 21 (FTP), 22 (SSH), 3389 (RDP), 445 (SMB)

dest_port	count
443	118
4444	33
23	29
21	28
22	28
3389	23
445	20

Those Allowed requests accessed these ports:

- 22 (SSH), 3389 (RDP), 4444, 21 (FTP), 23 (Telnet), 443 (HTTPS), 445 (SMB)

```
index=tryhackme sourcetype=Firewall
| rex "TCP\s+(?<src_ip>\d+\.\d+\.\d+\.\d+)"
| rex "(?<action>ALLOW|BLOCK)"
| rex "->\s+\d+\.\d+\.\d+\.\d+:(?<dest_port>\d+)"
| rex "->\s+(?<dest_ip>\d+\.\d+\.\d+\.\d+):(?<dest_port>\d+)"
| search src_ip="203.0.113.45"
| stats count by src_ip, dest_ip, dest_port, action
```

Time range: All time

✓ **297 events** (before 4/28/26 11:28:34.000 AM) No Event Sampling Job Smart Mode

Events Patterns **Statistics (35)** Visualization

Show: 50 Per Page Format Preview: On

src_ip	dest_ip	dest_port	action	count
203.0.113.45	10.0.0.20	21	BLOCK	25
203.0.113.45	10.0.0.20	22	BLOCK	22
203.0.113.45	10.0.0.20	23	ALLOW	1
203.0.113.45	10.0.0.20	23	BLOCK	21
203.0.113.45	10.0.0.20	3389	BLOCK	17
203.0.113.45	10.0.0.20	4444	ALLOW	2
203.0.113.45	10.0.0.20	4444	BLOCK	26
203.0.113.45	10.0.0.20	445	BLOCK	13
203.0.113.45	10.0.0.50	21	ALLOW	1
203.0.113.45	10.0.0.50	21	BLOCK	1
203.0.113.45	10.0.0.50	22	ALLOW	3
203.0.113.45	10.0.0.50	22	BLOCK	1
203.0.113.45	10.0.0.50	23	ALLOW	1

Interpretation: the same source IP (203.0.113.45) attempted connections to multiple hosts using different ports combinations. This behavior is consistent with:

- port scanning
- service probing

The presence of both blocked and allowed requests suggests that some services were accessible, while others were restricted by firewall rules.

b) **IP 203.0.113.10** – total events: 60 → 46 Blocked & 14 Allowed

```
index=tryhackme sourcetype=Firewall
| rex "TCP\s+(?<src_ip>\d+\.\d+\.\d+\.\d+)"
| rex "(?<action>ALLOW|BLOCK)"
| search src_ip="203.0.113.10"
| stats count by action
```

Time range: All time

✓ **60 events** (before 4/28/26 11:40:03.000 AM) No Event Sampling Job Smart Mode

Events Patterns **Statistics (2)** Visualization

Show: 50 Per Page Format Preview: On

action	count
ALLOW	14
BLOCK	46

Blocked requests targeted multiple ports:

- 3389 (RDP), 22 (SSH), 23 (Telnet), 4444, 445 (SMB), 21 (FTP)

```
index=tryhackme sourcetype=Firewall
| rex "TCP\s+(?<src_ip>\d+\.\d+\.\d+\.\d+)"
| rex "(?<action>ALLOW|BLOCK)"
| rex "->\s+\d+\.\d+\.\d+\.\d+:(?<dest_port>\d+)"
| search src_ip="203.0.113.10" action=BLOCK
| stats count by dest_port
```

Time range: All time

✓ **46 events** (before 4/28/26 11:40:49.000 AM) No Event Sampling Job Smart Mode

Events Patterns **Statistics (6)** Visualization

Show: 50 Per Page Format Preview: On

dest_port	count
3389	10
22	9
23	8
4444	7
445	7
21	5

Allowed requests targeted the same ports:

- 3389 (RDP), 22 (SSH), 23 (Telnet), 4444, 445 (SMB), 21 (FTP)

dest_port	count
23	4
4444	4
3389	2
445	2
21	1
22	1

Interpretation: This IP address shows a similar behavior pattern as the previous one. It attempted connections to multiple hosts using different port combinations, which is consistent with port scanning and service probing.

c) **IP 10.8.0.23** – this is a private IP address assigned via VPN

This IP address attempted to access the following ports: 445 (SMB), 22 (SSH), 3389 (RDP)

```
index=tryhackme sourcetype=Firewall
| rex "TCP\s+(?<src_ip>\d+\.\d+\.\d+\.\d+)"
| rex "(?<action>ALLOW|BLOCK)"
| rex "->\s+\d+\.\d+\.\d+\.\d+:(?<dest_port>\d+)"
| search src_ip="10.8.0.23" action=BLOCK
| stats count by dest_port
```

Time range: All time

✓ **26 events** (before 4/28/26 11:50:08.000 AM) No Event Sampling Job Smart Mode

Events Patterns **Statistics (3)** Visualization

Show: 50 Per Page Format Preview: On

dest_port	count
445	10
22	8
3389	8

It successfully accessed these ports: 22 (SSH), 445 (SMB), 3389 (RDP)

```
index=tryhackme sourcetype=Firewall
| rex "TCP\s+(?<src_ip>\d+\.\d+\.\d+\.\d+)"
| rex "(?<action>ALLOW|BLOCK)"
| rex "->\s+\d+\.\d+\.\d+\.\d+:(?<dest_port>\d+)"
| search src_ip="10.8.0.23" action=ALLOW
| stats count by dest_port
```

Time range: All time

✓ **64 events** (before 4/28/26 11:51:15.000 AM) No Event Sampling Job Smart Mode

Events Patterns **Statistics (3)** Visualization

Show: 50 Per Page Format Preview: On

dest_port	count
22	22
445	22
3389	20

Interpretation: This IP address was able to access multiple internal services, which suggests that it had valid access within the network.
From firewall logs alone, no further conclusion can be made.

2. VPN Correlation

Both external IP addresses successfully authenticated to the VPN.

a) **IP 203.0.113.45** – 27 successful logins → assigned 26 different internal IP addresses including 10.8.0.23

→ one internal IP address (10.8.0.23) was assigned twice

```
index=tryhackme sourcetype=vpn
| rex "\d{2}:\d{2}:\d{2}\s+(?<src_ip>\d+\.\d+\.\d+\.\d+)"
| search src_ip="203.0.113.45"
| stats count by assigned_ip
```

Time range: All time

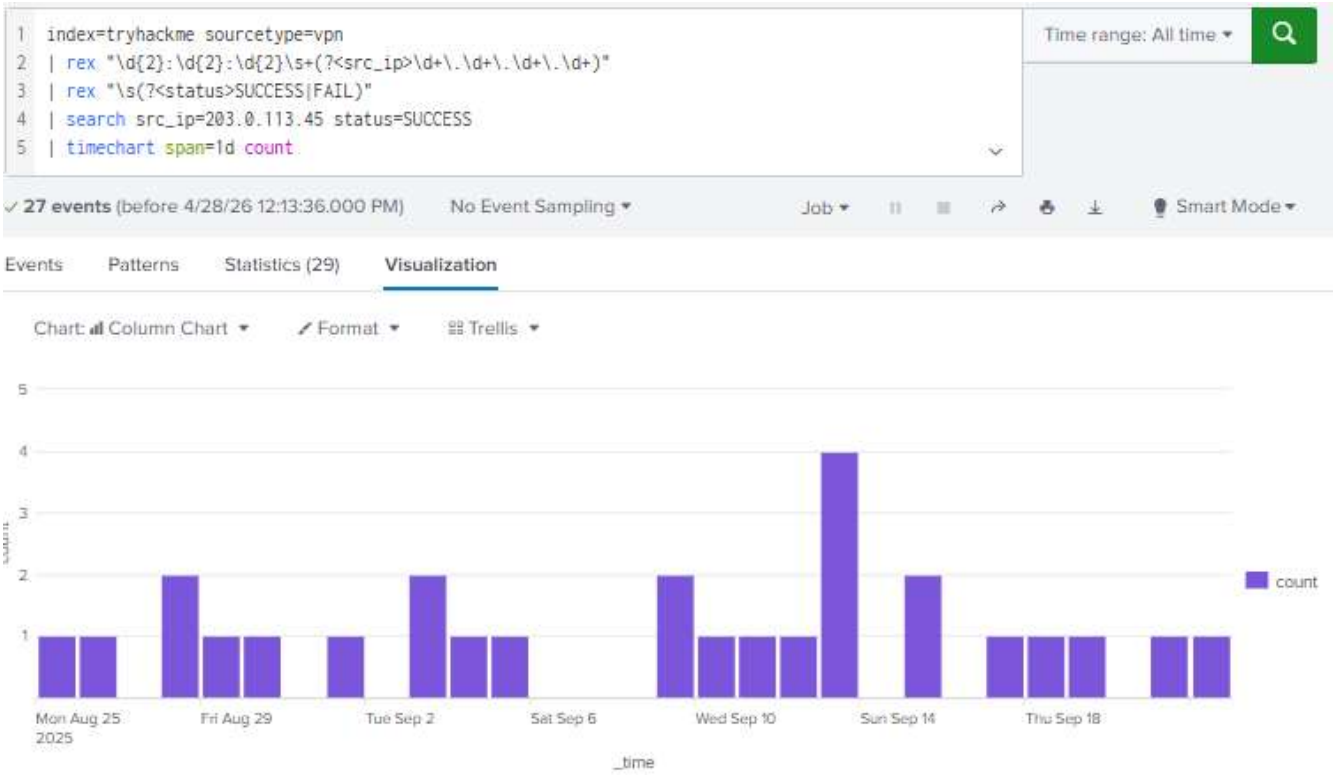
✓ **145 events** (before 4/28/26 11:58:53.000 AM) No Event Sampling Job Smart Mode

Events Patterns **Statistics (26)** Visualization

Show: 50 Per Page Format Preview: On

assigned_ip	count
10.8.0.23	2
10.8.0.104	1
10.8.0.118	1
10.8.0.12	1
10.8.0.126	1
10.8.0.131	1
10.8.0.134	1
10.8.0.135	1
10.8.0.143	1
10.8.0.148	1
10.8.0.161	1
10.8.0.172	1
10.8.0.181	1
10.8.0.187	1

The logins were irregular, without a clear pattern. The highest activity was observed on Sep-13, 2025 (4 logins).



Interpretation: This suggests repeated VPN access with new sessions being created over time, each receiving different internal IP address.
The behavior appears unusual due to the high number of logins and assigned Ips. However, the lack of a clear pattern makes it difficult to determine whether the activity was fully automated.

b) **IP 203.0.113.10** - 15 successful logins → assigned 15 different internal IP addresses, including 10.8.0.23

```
index=tryhackme sourcetype=vpn
| rex "\d{2}:\d{2}:\d{2}\s+(?<src_ip>\d+\.\d+\.\d+\.\d+)"
| search src_ip="203.0.113.10"
| stats count by assigned_ip
```

Time range: All time

✓ **15 events** (before 4/28/26 12:23:44.000 PM) No Event Sampling Job Smart Mode

Events Patterns **Statistics (15)** Visualization

Show: 50 Per Page Format Preview: On

assigned_ip	count
10.8.0.14	1
10.8.0.23	1
10.8.0.31	1
10.8.0.32	1
10.8.0.42	1
10.8.0.63	1
10.8.0.83	1
10.8.0.85	1
10.8.0.86	1
10.8.0.114	1

The logins were irregular, without a clear pattern and without any significant spike.

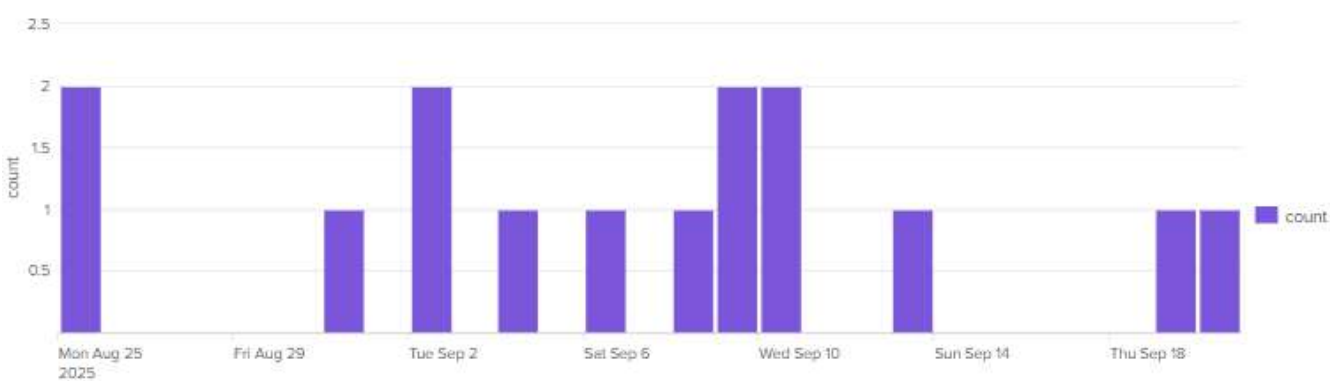
```
index=tryhackme sourcetype=vpn
| rex "\d{2}:\d{2}:\d{2}\s+(?<src_ip>\d+\.\d+\.\d+\.\d+)"
| rex "\s(?<status>SUCCESS|FAIL)"
| search src_ip=203.0.113.10 status=SUCCESS
| timechart span=1d count
```

Time range: All time

✓ **15 events** (before 4/28/26 12:23:01.000 PM) No Event Sampling Job Smart Mode

Events Patterns Statistics (27) **Visualization**

Chart: Column Chart Format Trellis



Interpretation: this shows a similar pattern as the previous IP address: repeated VPN logins, new sessions with different internal IP addresses.

VPN Users Overview

During the analyzed period, 4 users were observed:

- svc_backup – logged 137 times
- jsmith – logged 30 times
- bob – logged 29 times
- alice – logged 24 times

Failed login Attempts

- svc_backup – 118 failed attempts
- jsmith – 2 failed attempts
- alice – 1 failed attempt
- bob – 0 failed attempt

```
index=tryhackme sourcetype=vpn
| rex "\d{2}:\d{2}:\d{2}\s+\d+\.\d+\.\d+\.\d+\s+(?<user>\w+)\s+(?<status>\w+)"
| search status=FAIL
| stats count by user
| sort -count
```

Time range: All time

✓ 121 events (before 4/28/26 12:37:33.000 PM) No Event Sampling Job Smart Mode

Events Patterns Statistics (3) Visualization

Show: 50 Per Page Format Preview: On

user	count
svc_backup	118
jsmith	2
alice	1

The number of failed attempts for svc_backup is significantly higher compared to the other users.

The account svc_backup was identified as suspicious due to:

- a high number of login attempts
- a high number of failed logins
- logins from multiple public IP addresses
- no clear time pattern

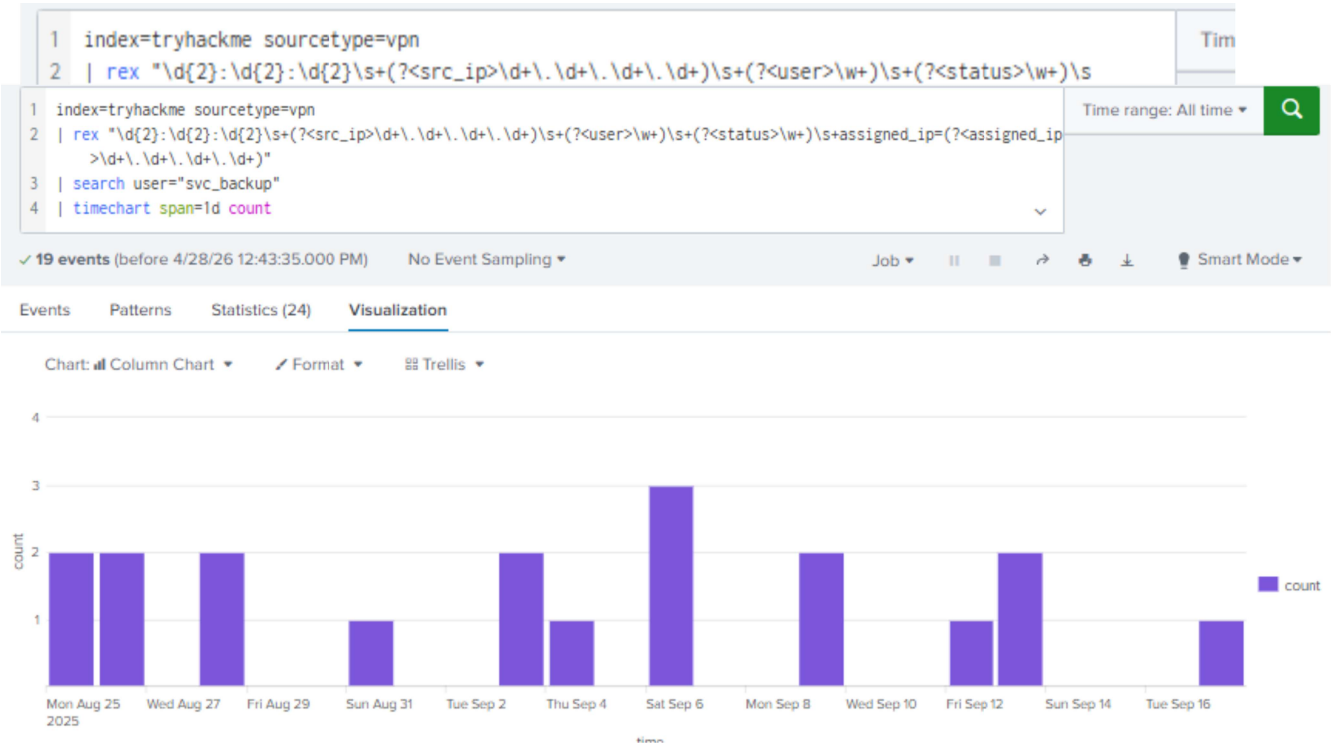
This behavior is not typical for a service account, which would normally:

- authenticate from a limited number of IP addresses
- show a more consistent pattern

Interpretation: This activity may indicate:

- a possible brute force attempt
- or a potentially compromised service account

The exact reason for failed login attempts cannot be determined from available logs. Additional authentication logs would be required for deeper analysis.



3. IDS Analysis/Internal Activity

This section summarizes the most relevant suspicious IP addresses identified in IDS logs, along with their associated signatures.

```
index=tryhackme sourcetype=IDS
| rex "\[\*\*\]\s+\[\d+:\d+:\d+\]\s+(?<signature>.*?)\s+\[\*\*\]"
| rex "\{TCP\}\s+(?<src_ip>\d+\.\d+\.\d+\.\d+)"
| stats count by signature src_ip
| sort -count
```

Time range: All time

✓ **983 events** (before 4/28/26 1:09:55.000 PM) No Event Sampling Job Smart Mode

Events Patterns **Statistics (28)** Visualization

Show: 50 Per Page Format Preview: On

signature	src_ip	count
ET POLICY VPN Authentication Failure	203.0.113.45	120
ET SCAN Possible Portscan	203.0.113.45	120
ET TROJAN Possible C2 Beaconing	10.0.0.60	80
ET INFO Possible HTTP POST Large Upload	10.0.0.51	60
ET INFO Possible Benign Scan	198.51.100.45	38
ET EXPLOIT Possible MS-SMB Lateral Movement	10.8.0.23	32
ET INFO Possible Benign Scan	203.0.113.45	32
ET SCAN Possible SSH Scan	10.8.0.23	30
ET POLICY Suspicious HTTP	192.0.2.115	29
ET WEB_SERVER Possible SQL Injection	203.0.113.100	29
ET EXPLOIT Possible RDP Brute Force	10.8.0.23	28
ET INFO Possible Benign Scan	198.51.100.92	28
ET POLICY Suspicious HTTP	203.0.113.100	26
ET WEB_SERVER Possible SQL Injection	192.0.2.115	26
ET WEB_SERVER Possible SQL Injection	203.0.113.10	26
ET POLICY Suspicious HTTP	203.0.113.10	25
ET POLICY Suspicious HTTP	198.51.100.92	24
ET WEB_SERVER Possible SQL Injection	198.51.100.45	24
ET WEB_SERVER Possible SQL Injection	203.0.113.45	24
ET POLICY Suspicious HTTP	198.51.100.77	23
ET POLICY Suspicious HTTP	203.0.113.45	23
ET INFO Possible Benign Scan	192.0.2.115	22
ET INFO Possible Benign Scan	203.0.113.100	22
ET INFO Possible Benign Scan	203.0.113.10	20
ET INFO Possible Benign Scan	198.51.100.77	19
ET WEB_SERVER Possible SQL Injection	198.51.100.77	19
ET WEB_SERVER Possible SQL Injection	198.51.100.92	19
ET POLICY Suspicious HTTP	198.51.100.45	15

1) **IP 10.8.0.23** (VPN-assigned)

IDS logs show that IP generated 90 events on Sept-05, 2025, approximately 6 events every hour
Detected activity:

- SSH scanning

- RDP brute force attempts
- SMB lateral movement (port 445)

```
index=tryhackme sourcetype=IDS
| rex "\[\*\*\]\s+\[\d+:\d+:\d+\]\s+(?<signature>.*?)\s+\[\*\*\]"
| rex "\{TCP\}\s+(?<src_ip>\d+\.\d+\.\d+\.\d+)"
| search src_ip=10.8.0.23
| stats count by signature
| sort -count
```

Time range: All time

✓ **90 events** (before 4/28/26 1:06:20.000 PM) No Event Sampling Job Smart Mode

Events Patterns **Statistics (3)** Visualization

Show: 50 Per Page Format Preview: On

signature	count
ET EXPLOIT Possible MS-SMB Lateral Movement	32
ET SCAN Possible SSH Scan	30
ET EXPLOIT Possible RDP Brute Force	28

Interpretation:

- internal reconnaissance
- attempts to move within the network

```
index=tryhackme sourcetype=IDS | rex "\]\s+(?<signature>.*?)\s+\[\*\*\]"
| rex "\{TCP\}\s+(?<src_ip>\d+\.\d+\.\d+\.\d+)"
| search src_ip="10.8.0.23"
| timechart span=1d count
```

Time range: All time

✓ **90 events** (before 4/28/26 1:03:13.000 PM) No Event Sampling Job Smart Mode

Events Patterns **Statistics (1)** Visualization

Show: 50 Per Page Format Preview: On

_time	count
2025-09-05	90

2) **203.0.113.46**

Detected Activity:

- VPN Authentication Failure
- Possible Portscan

This behavior was already observed in firewall logs and is confirmed by IDS

3) **IP 10.0.0.60**

IDS detected

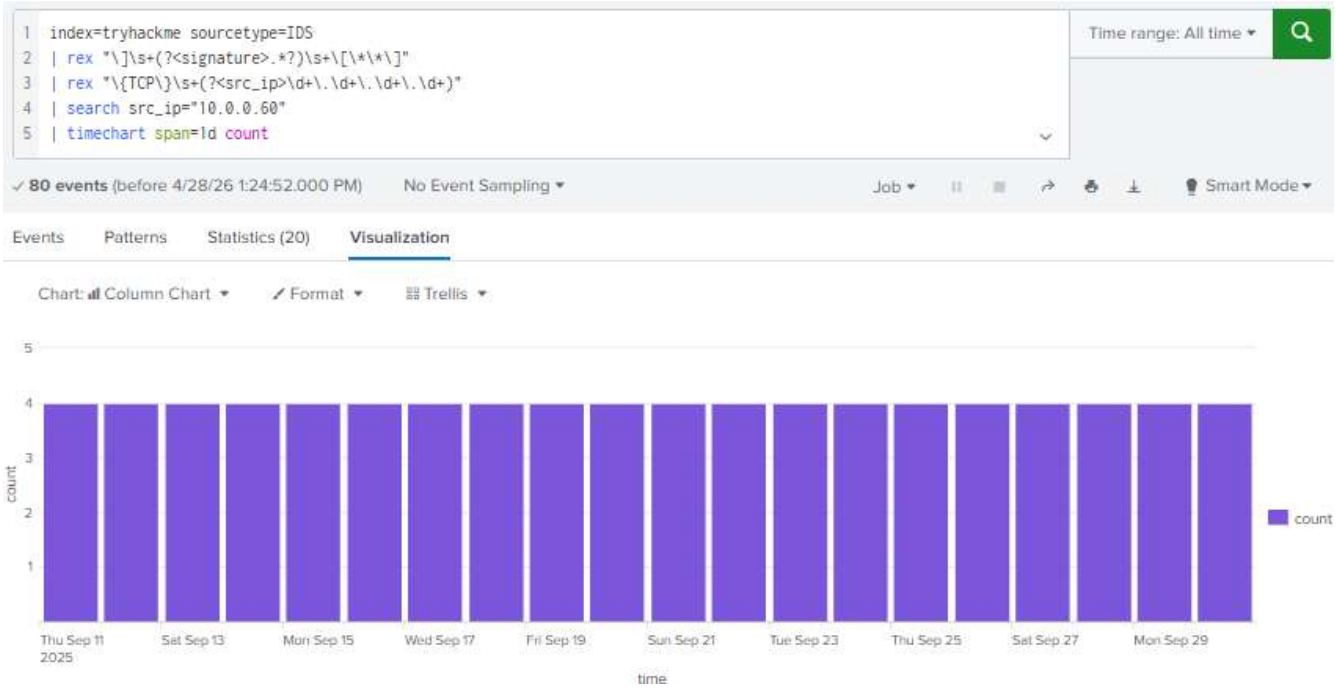
- ET Trojan Possible C2 Beaconing

Observed behavior:

- communication from internal host 10.0.0.60 to external IP 198.51.100.77
- activity between Sep-11, 2025 – Sep-30, 2025
- approximately 4 events per day

Interpretation:

- persistent outbound communication (198.51.100.77)
- possible compromised host communicating with a command-and-control (C2) server



4) **IP 10.0.0.51**

IDS detected:

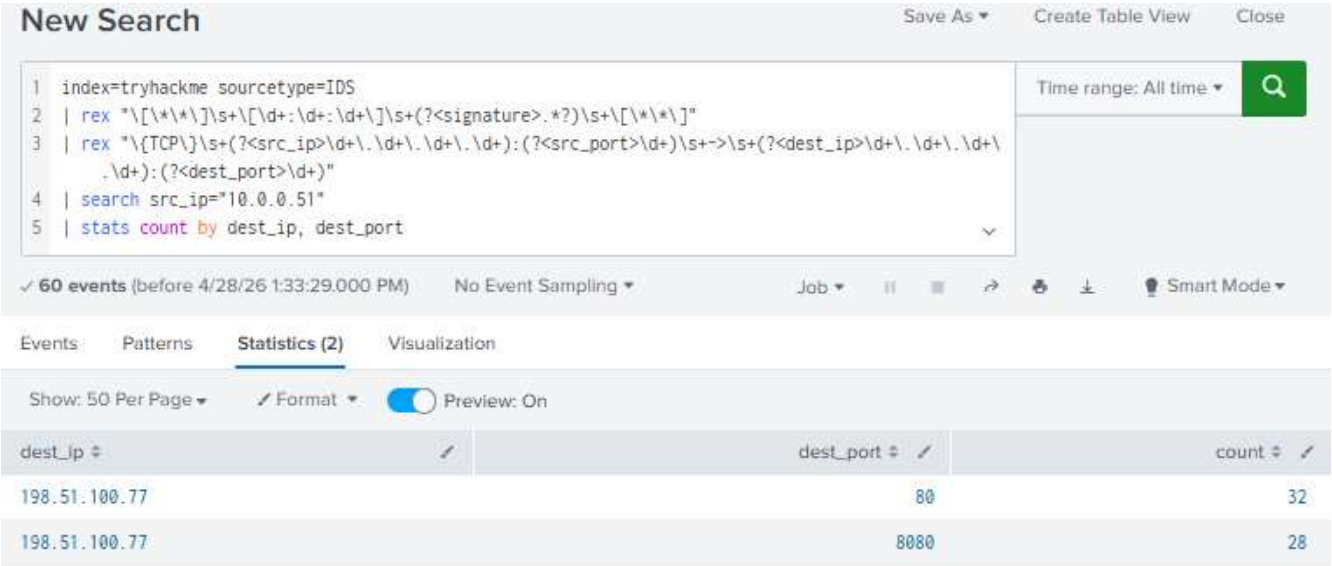
- HTTP POST Large Upload

Observed behavior:

- communication to external IP 198.51.100.77
- ports 80 and 8080

Interpretation:

- possible data transfer
- potential data exfiltration (not confirmed, would need additional logs)



Conclusion:

The analysis identified multiple types of suspicious activity across firewall, VPN, and IDS logs.

External IP addresses (203.0.113.45 & 203.0.113.10) showed behavior consistent with port scanning and successfully authenticated to the VPN multiple times, receiving different internal IP addresses.

One of the VPN-assigned IP (10.8.0.23) performed internal scanning, brute force attempt, and SMB lateral movement, indicating attempts to explore and access internal systems.

A separate internal host (10.0.0.60) exhibited persistent outbound communication identified as C2 beaconing, which may indicate a compromised system communicating with an external server (198.51.100.77).

Additionally, another internal host (10.0.0.51) showed signs of possible data transfer via HTTP POST requests, although data exfiltration cannot be confirmed from available logs.

An unusual pattern was also observed for the service account svc_backup, which showed a high number of login attempts (both successful and failed) from multiple public IP addresses, suggesting potential misuse or compromise of the account.

Overall, the logs indicate a sequence of suspicious activities including external reconnaissance, repeated VPN access, internal scanning, and persistent outbound communication.

However, due to limited visibility, it is not possible to confirm the exact point of compromise or definitively link all observed activities together.